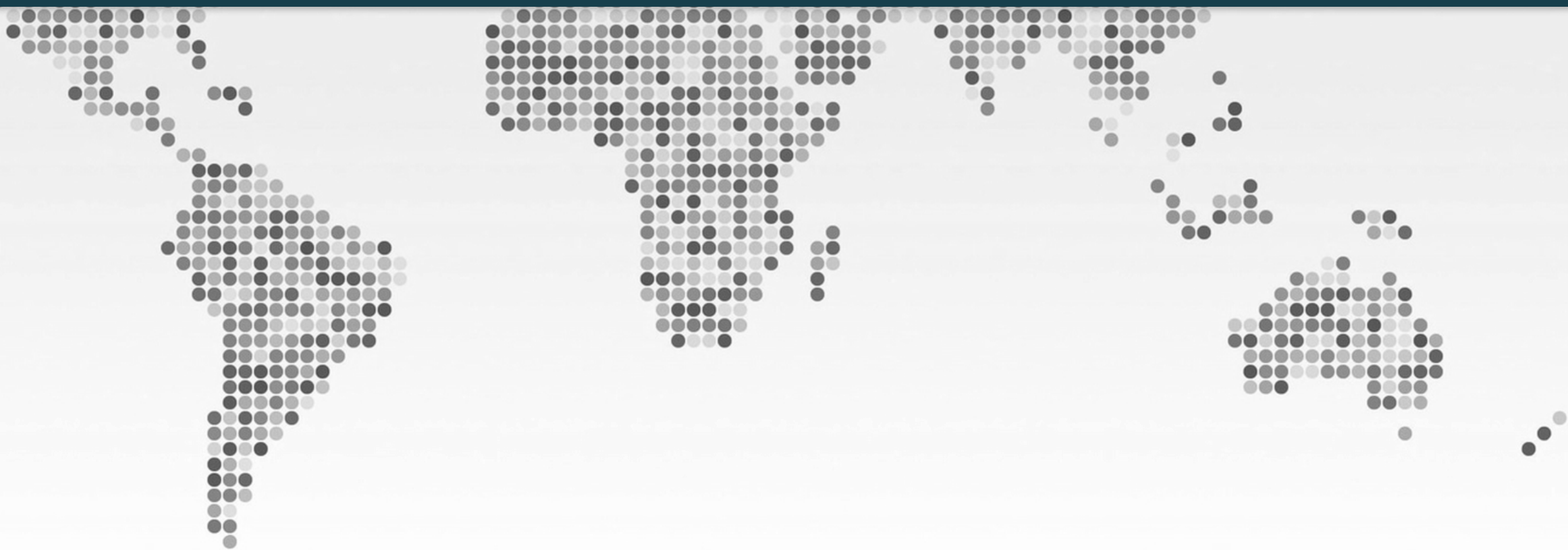


Software Engineering for Artificial Intelligence

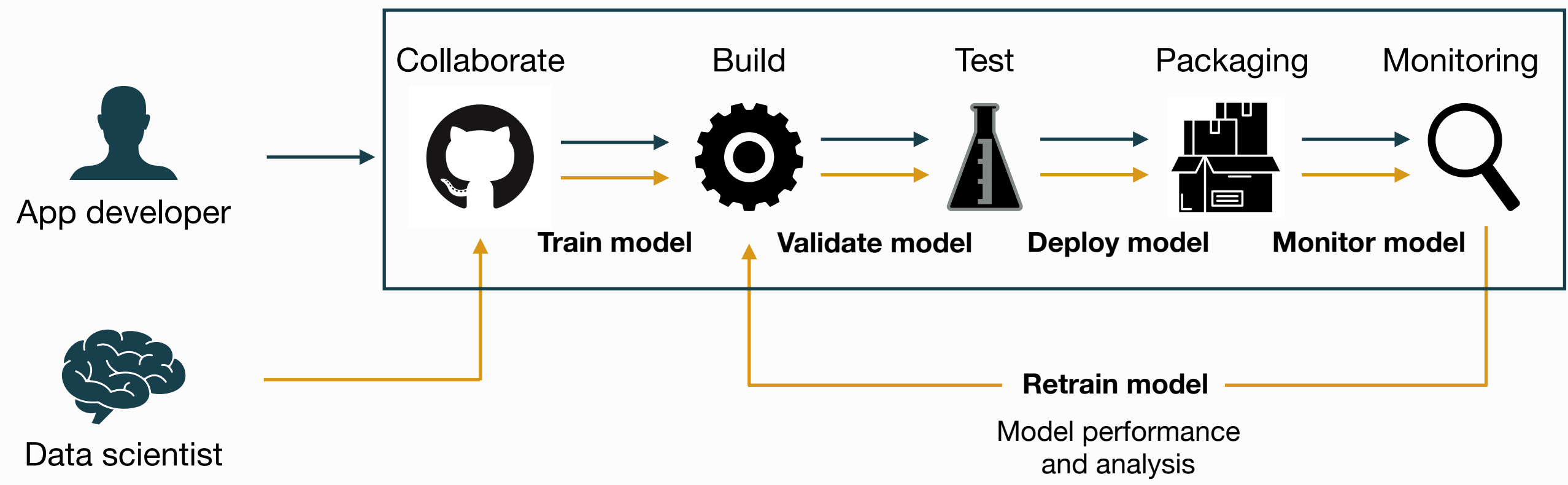
Midterm In-Vivo Quiz



Software Engineering for Artificial Intelligence: Introduction

Engineering Artificial Intelligence Systems: From DevOps to MLOps

Q1. Define The Activities You Typically Do in MLOps and How SE4AI can Help with that



Model reproducibility	✓	Model validation	✓	Model deployment	✓	Model retraining	✓
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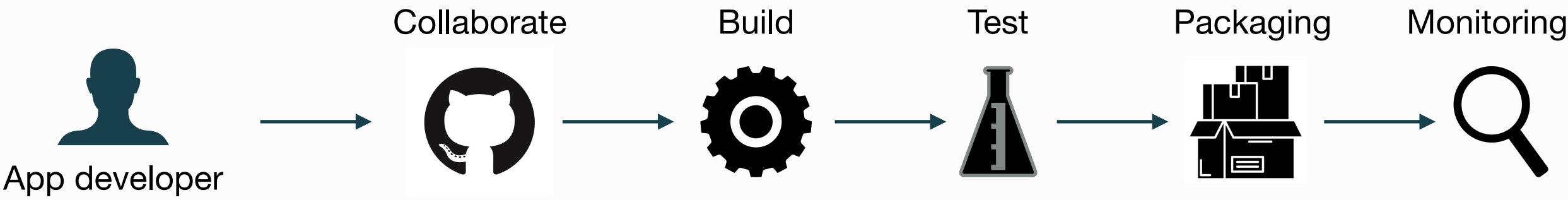
What we are going to further investigate in the next few weeks pertain to MLOps and how to ensure the properties of the artificial intelligence models built within complex software systems.

Our view on the deployment of the models will be, however, limited by the time available. Advanced topics on ML deployment, e.g., federated machine learning, are out of scope.

Software Engineering for Artificial Intelligence: Introduction

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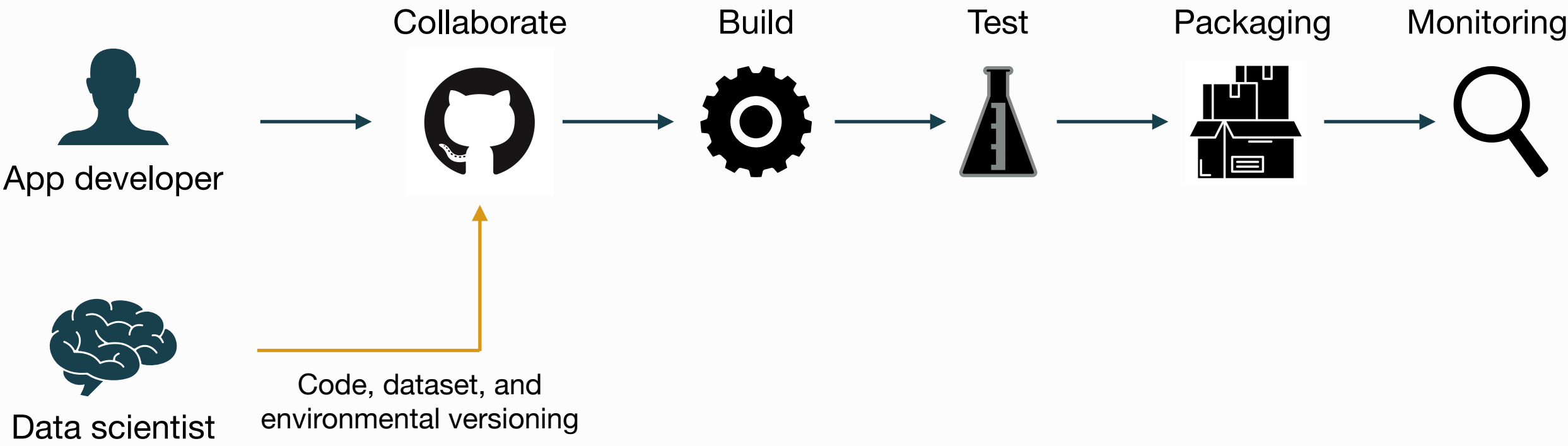


Model reproducibility	✗	Model validation	✗	Model deployment	✗	Model retraining	✗
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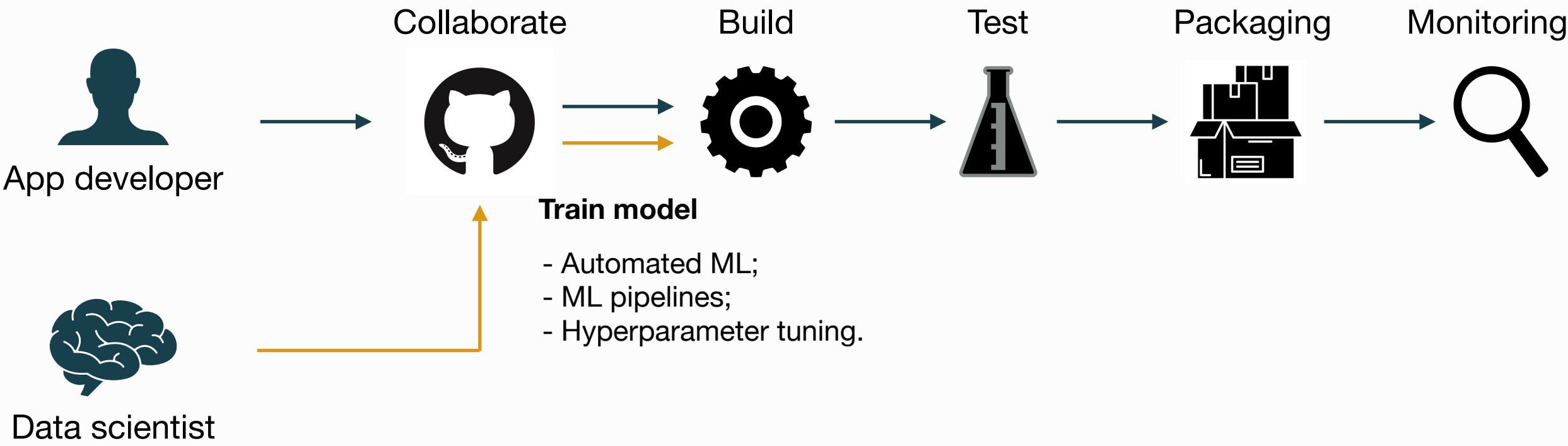


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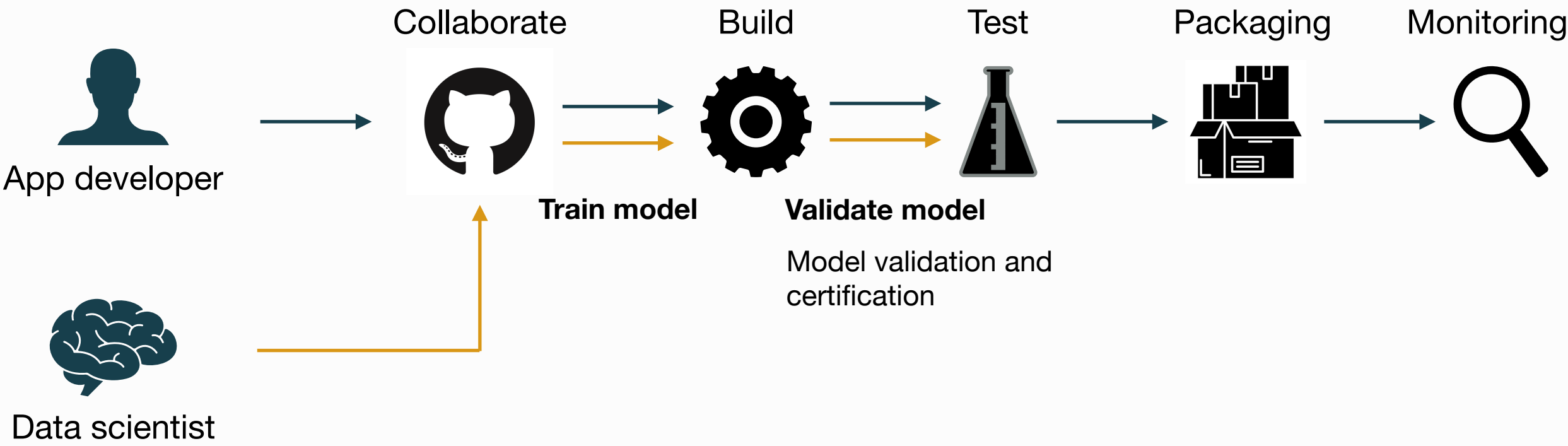


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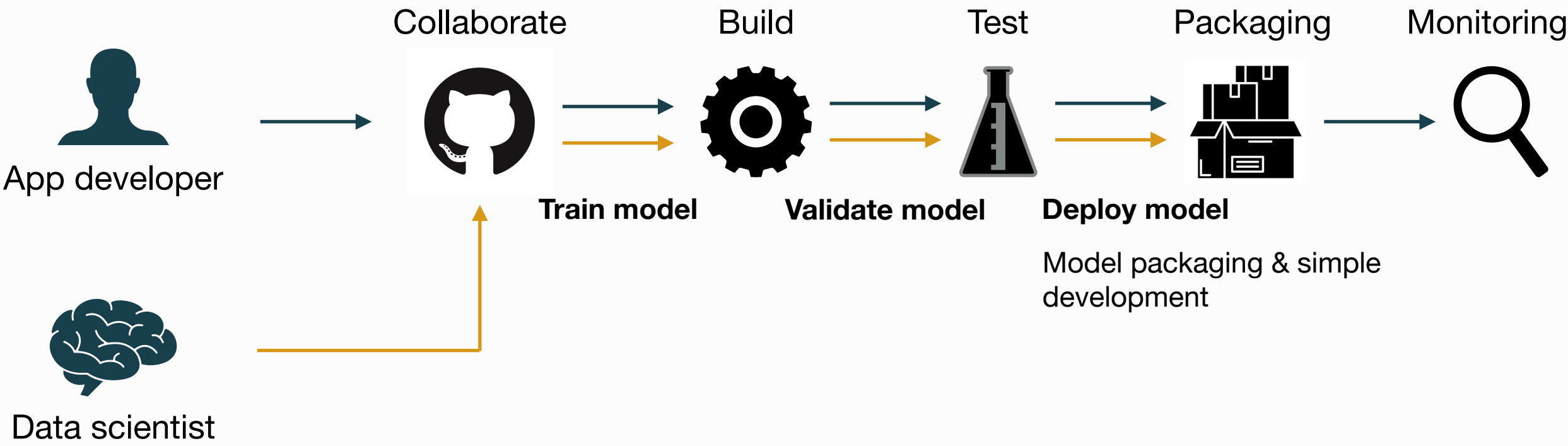


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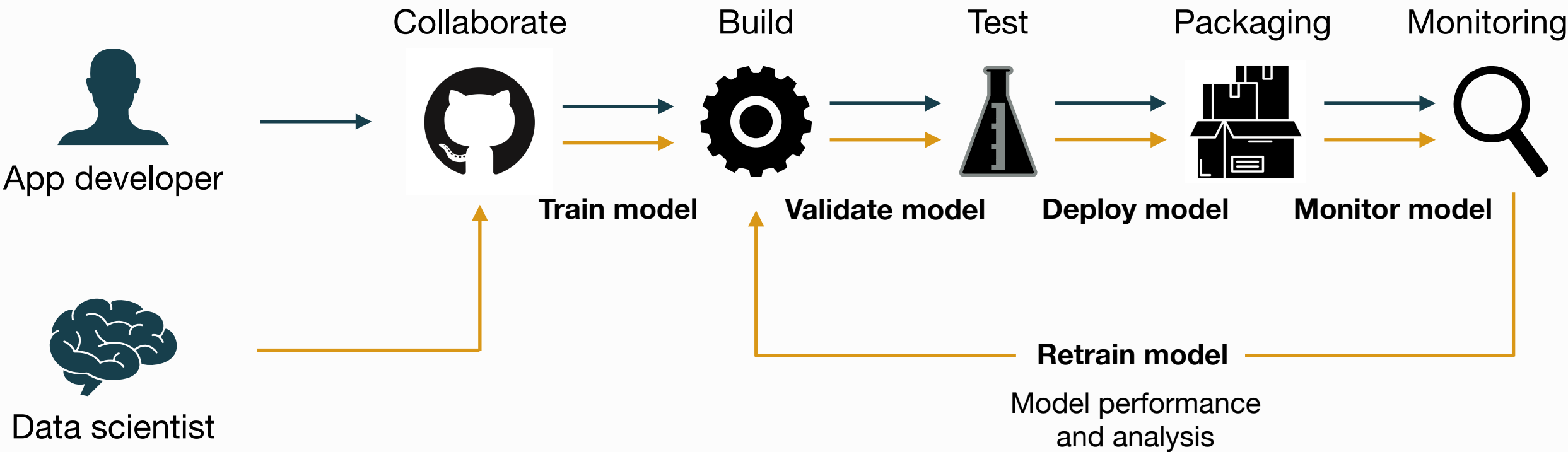


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Software Engineering for Artificial Intelligence: Introduction

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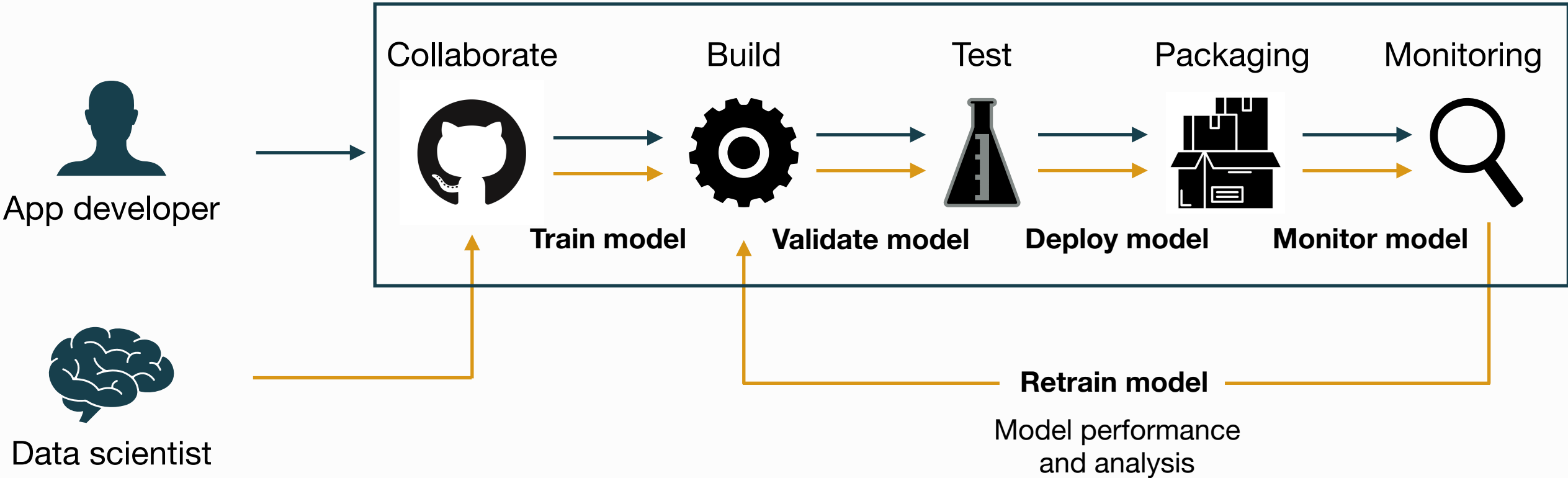


Model reproducibility	✓	Model validation	✓	Model deployment	✓	Model retraining	✓
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Software Engineering for Artificial Intelligence: Introduction

Engineering Artificial Intelligence Systems: From DevOps to MLOps

Q1. Define The Activities You Typically Do in MLOps and How SE4AI can Help with that



**Model
reproducibility** ✓

**Model
validation** ✓

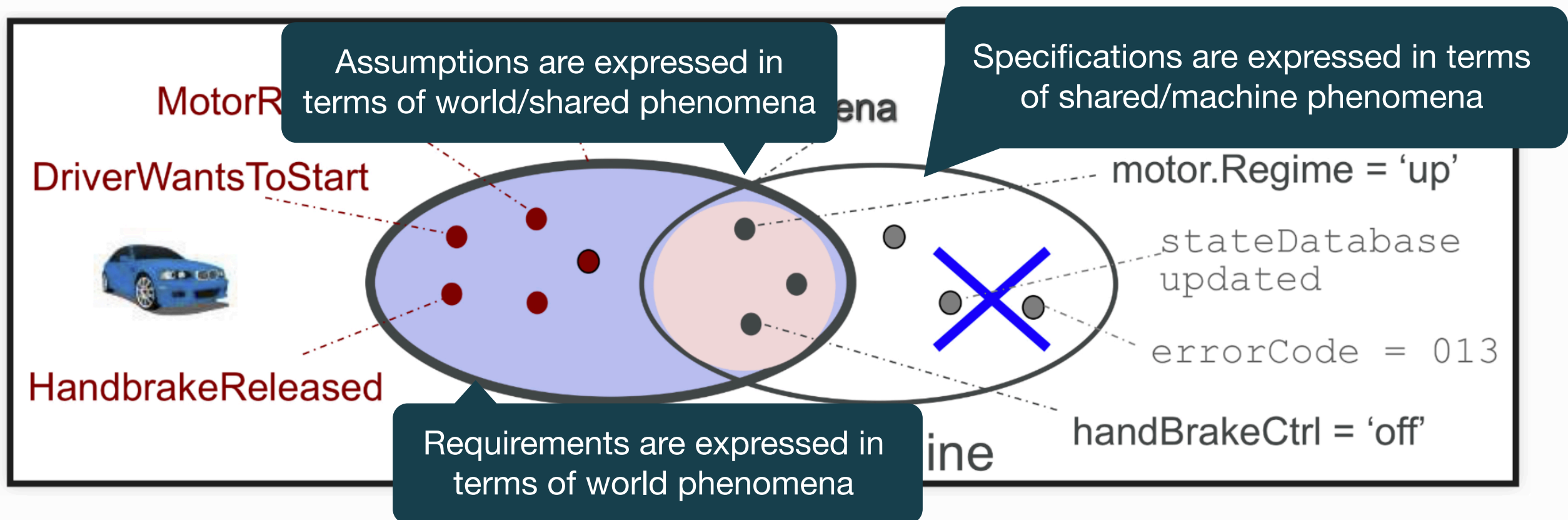
**Model
deployment** ✓

**Model
retraining** ✓

Software Engineering for Artificial Intelligence: Requirements Engineering

Requirements Engineering: Implications for AI-intensive systems

Q2. What is the difference between World, Shared, and Machine Phenomena in RE?



Shared phenomena: Interface between the world and the machine (i.e., the software).

Requirements model stakeholder's needs and are expressed in natural language in order to be understood by all interested parts (e.g., "*the driver wants to start*").

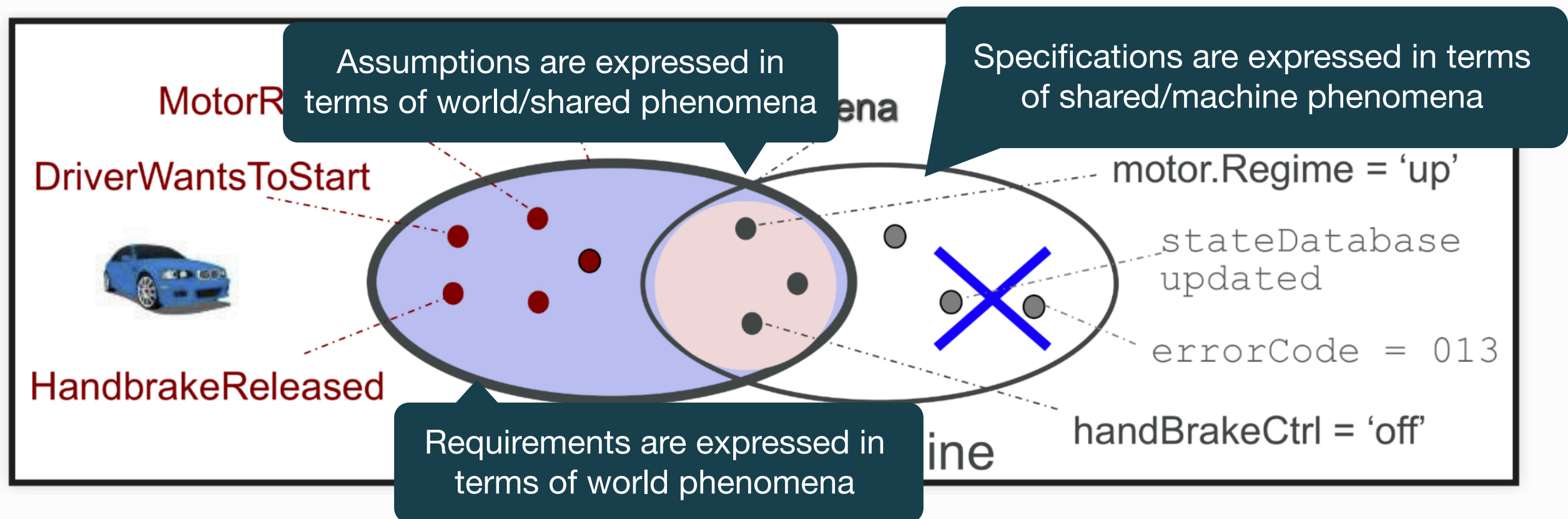
Assumptions are needed to abstract the surrounding world and must necessarily consider the way customers interact with the system (e.g., `motor.Regime = 'up'`).

Requirement specifications map how the system reacts to requirements and assumptions (e.g., `stateDatabase updated`).

Software Engineering for Artificial Intelligence: Requirements Engineering

Requirements Engineering: Implications for AI-intensive systems

Q2. What is the difference between World, Shared, and Machine Phenomena in RE?



software).

uage in order

sarily

= 'up').

d

What could go wrong?

- (1) Missing/Incorrect environmental assumptions;
- (2) Wrong specifications;
- (3) Inconsistency in assumptions and specifications;
- (4) Inconsistency in requirements.

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Requirements Engineering: Implications for AI-intensive systems

Q3. What is the Difference between Data and Feature?

Suppose that you would like to design an automated solution able to forecast the number of students who will attend the next SE4AI lecture.

Which data would you use?

Raw data coming from the MS Teams reports

MS Teams collects statistics on the student's attendance for each lecture scheduled. This sounds like the ideal dataset to use! But is this the only available data?

Raw data coming from the course/exam schedule

Attendance might depend on other factors that you can measure, like the co-occurrence of institutional events. Raw data can be extracted by mining EasyLesson!

Which feature(s) would you build?

Average number of students per lecture?

Number of students attending the last lecture?

Presence of other institutional events?

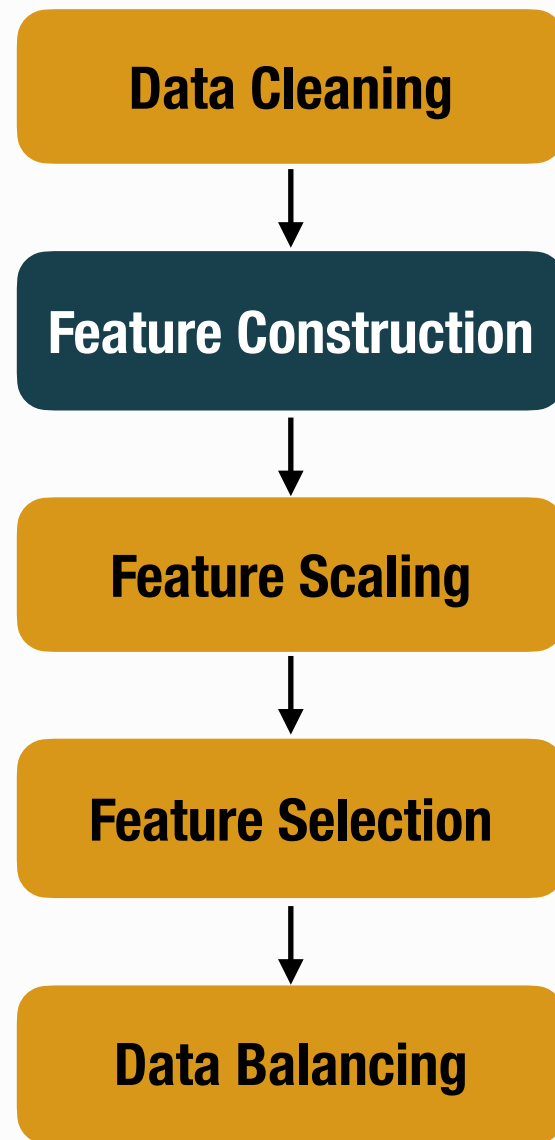
Additional factors coming from your knowledge?

In other words, feature engineering is a creative process that involves the combination of data elaboration and domain knowledge!

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

Q3. What are the tasks applied in a Typical Pre-processing pipeline?

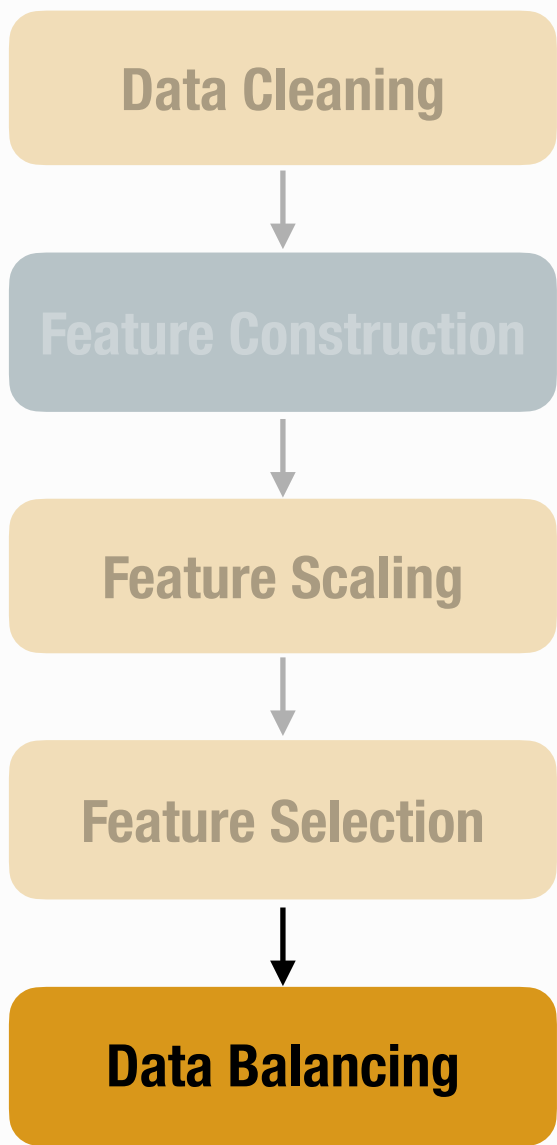


No answer for that... otherwise, what am I supposed to ask at the exam?!

Software Engineering for Artificial Intelligence: Data and Feature Engineering

Data and Feature Engineering: Toward Data Quality and Feature Engineering

Q4. What is SMOTE? And how does it work?



Data Balancing. Set of techniques that might convert an unbalanced dataset into a balanced one by means of analysis of the data available.

- Select a random instance X of the minority class.
- Find the k -nearest neighbors of X , e.g., 4.
- Randomly choice of the the k -nearest neighbors of X .
- Generate a synthetic instance.
- Repeat for all neighbors.

X_amount	X_ratio
2800	0.79

X4_amount	X4_ratio
2770	0.94

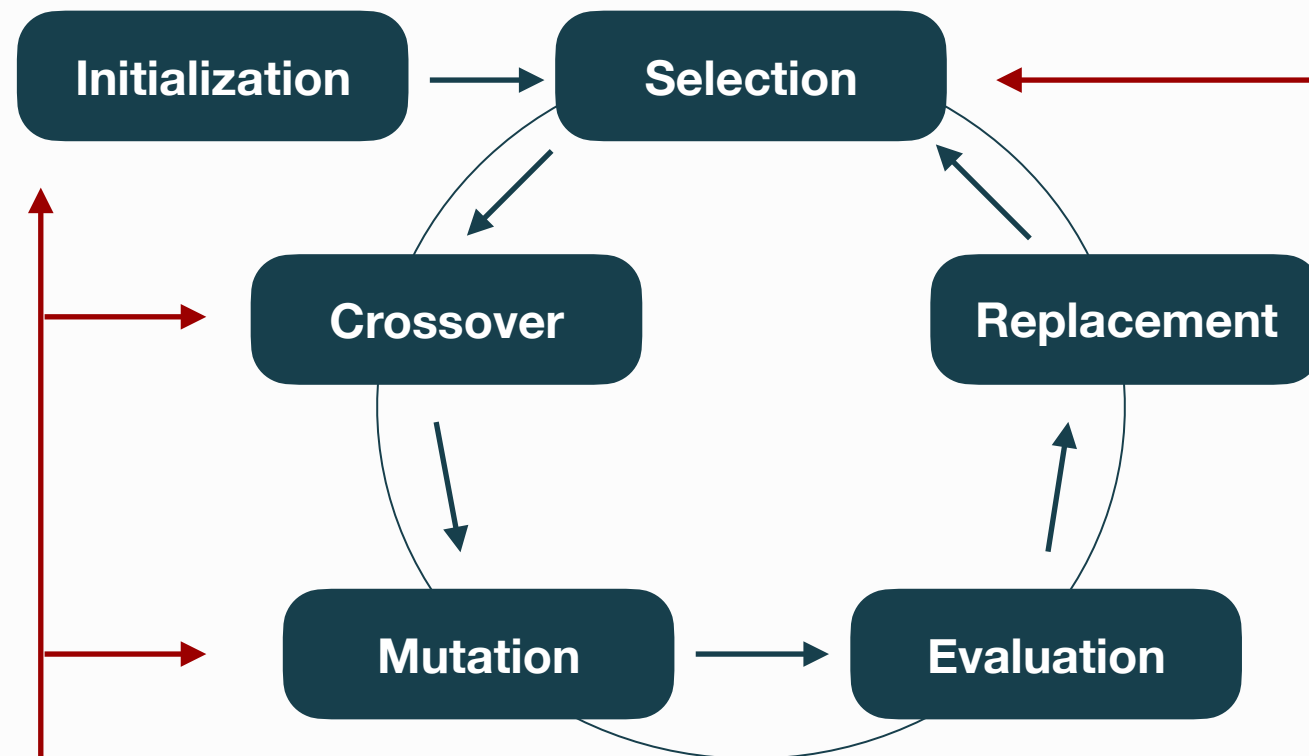
Choose random number in $[0;1]$, e.g., 0.6.
Scaling of the data based on this number.

$2.800 + 0.6 * (2770 - 2800)$	$0.79 + 0.6 * (0.94 - 0.79)$
Xnew_amount	Xnew_ratio
2782	0.88

Software Engineering for Artificial Intelligence: Configuring AI Solutions I

Configuring Artificial Intelligence Solutions: Search-Based Algorithms - Exercise

Q5. Define the methodology to create a single-objective GA to configure a machine learning model of your choice, e.g., a Random Forest?



Any machine learning algorithm might be evaluated against its accuracy, right?

So, you might want to consider the evaluation metric that best fits the problem of interest.

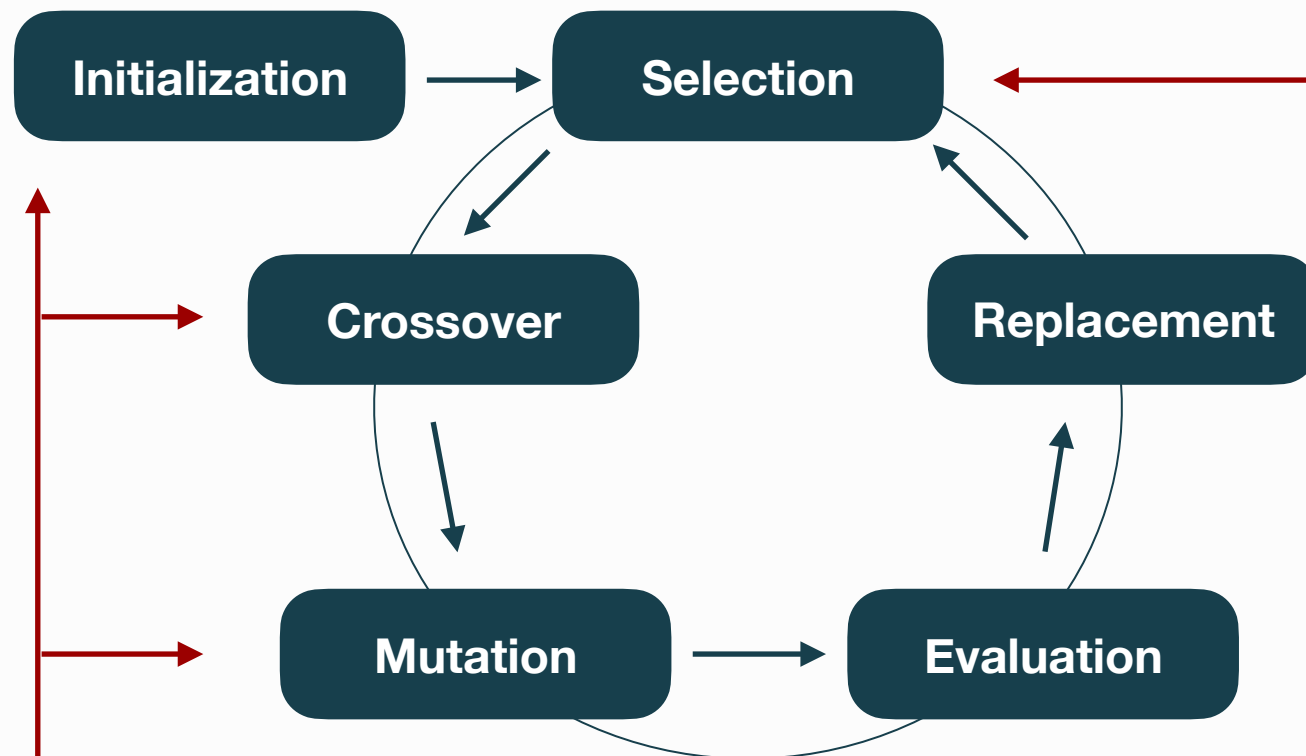
A chromosome might be a configuration value. A gene is a value of a certain hyper-parameter, a chromosome a composition of values to hyper-parameters.

Crossover may “simply” vary values of multiple hyper-parameter configurations, while mutations might assume any random values of the search space.

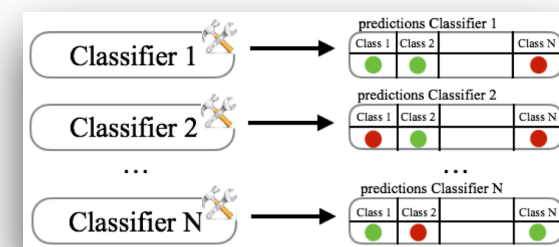
Software Engineering for Artificial Intelligence: Configuring AI Solutions I

Configuring Artificial Intelligence Solutions: Search-Based Algorithms - Exercise

Q6. Define the methodology to create a multi-objective GA to ensemble machine learning algorithms, i.e., how to use GAs to create effective ensemble learning methods?



What makes an ensemble suitable?
Orthogonality, right?



So, we evolve solutions that maximize the orthogonality of the learners.

But we might want to consider
secondary objectives, e.g., time?

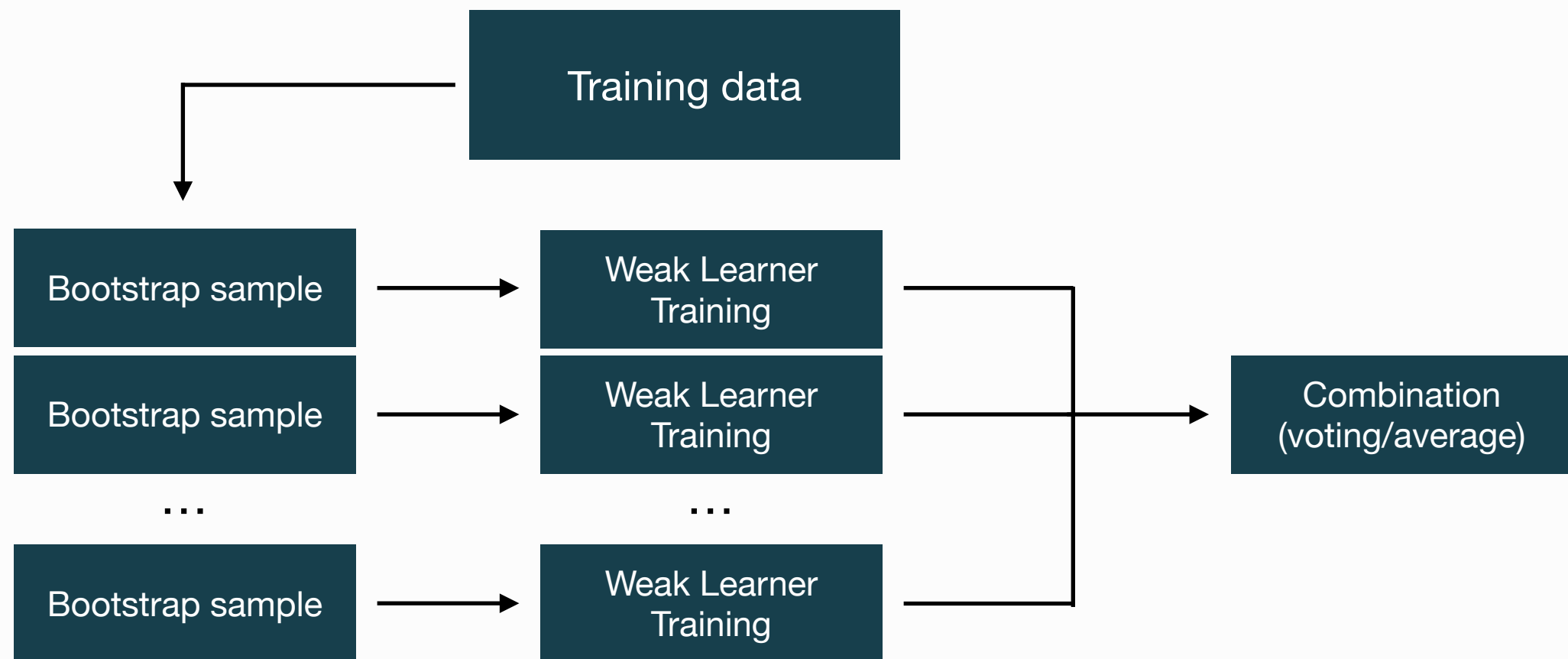
The representation must be a viable solution! Hence: the problem can be seen as a selection of the best learns for a given problem, right?! In other terms, guess what, it is yet another binary array representation!

Crossover and mutation operations might be similar to what we discussed during the lectures, unless you want to create ad-hoc operators.

Configuring Artificial Intelligence Solutions: Advanced

Q7. What is the difference between bagging and stacking ensemble learning?

Bagging. This comes from the abbreviation of Bootstrap AGGREGatING and typically involves the use of a single weak learner trained with different samples of the same training dataset—the Random Forest is a special case of bagging.



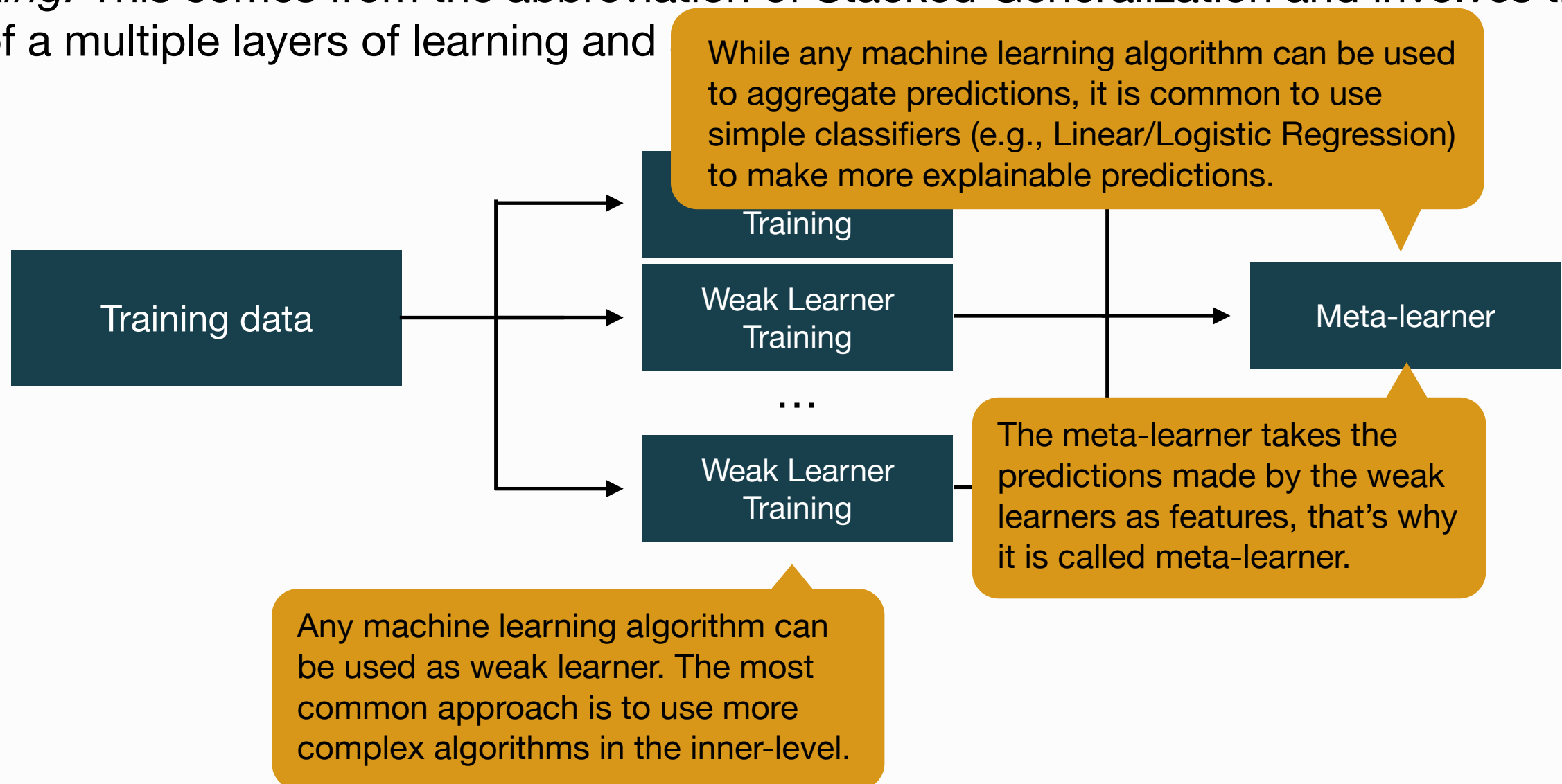
Software Engineering for Artificial Intelligence: Configuring AI Solutions I

Configuring Artificial Intelligence Solutions: Advanced

Q7. What is the difference between bagging and stacking ensemble learning?

Bagging. This comes from the abbreviation of Bootstrap AGGREGatING and typically involves the use of a single weak learner trained with different samples of the same training dataset—the Random Forest is a special case of bagging.

Stacking. This comes from the abbreviation of Stacked Generalization and involves the use of a multiple layers of learning and



Configuring Artificial Intelligence Solutions: Advanced

Q8. How would you configure a deep learner given this distribution of data?



Software Engineering for Artificial Intelligence: Configuring AI Solutions II

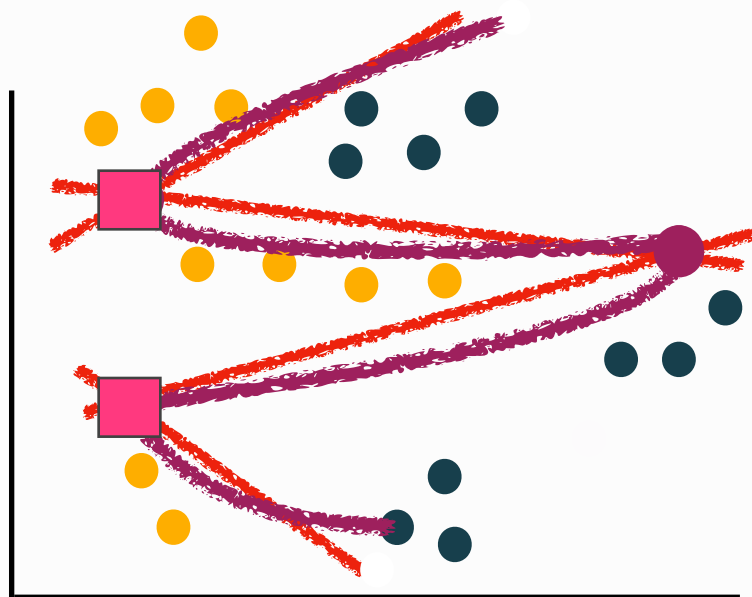
Configuring Artificial Intelligence Solutions: Advanced

Deep Learning

- Neural Networks;
- Convolutional Neural Networks;
- Recurrent Neural Networks.

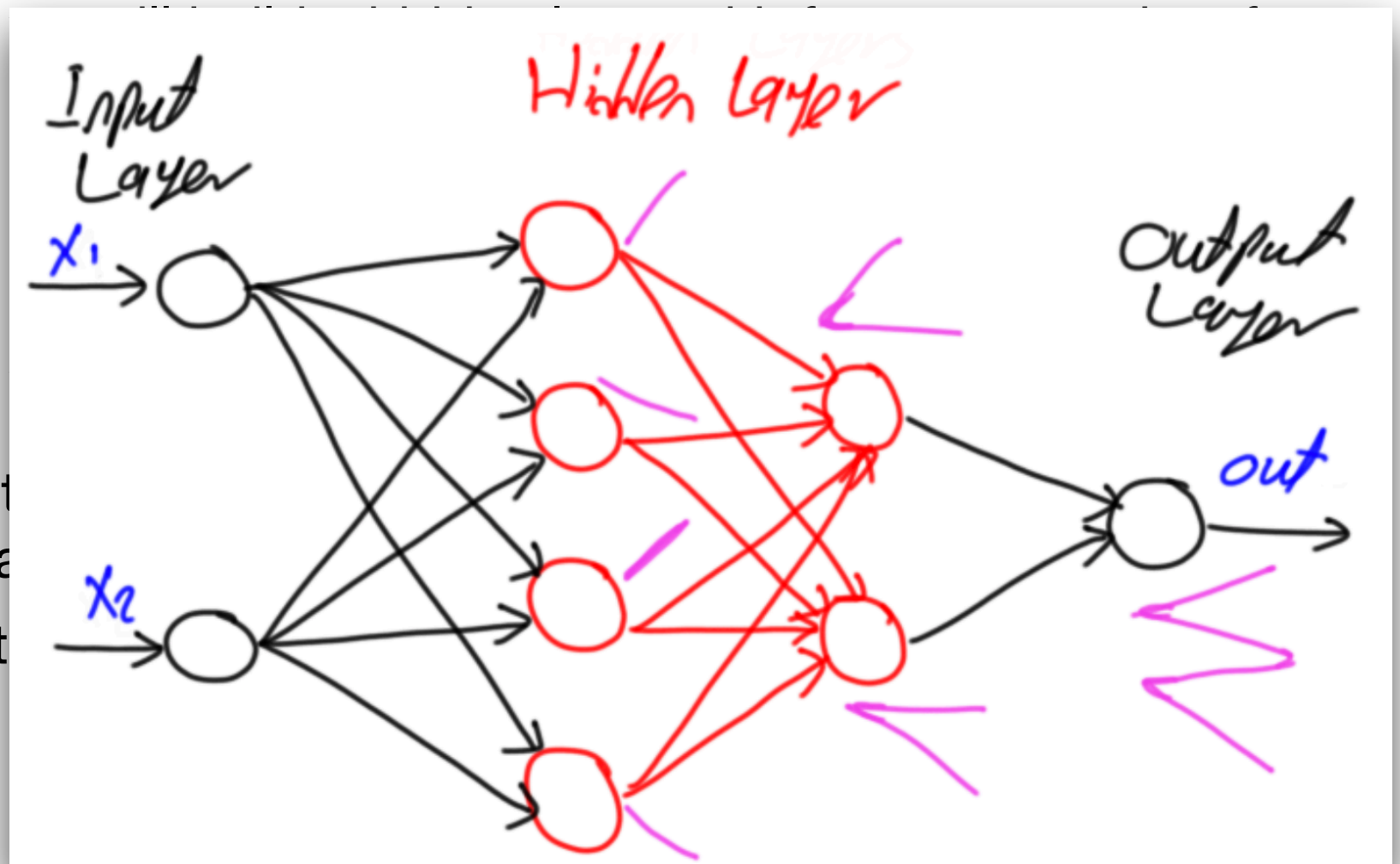
Neural Networks. Machine learning model inspired by the networks of biological neurons.

Easy, right? Let's do another example.



Graphically speaking, the addition of the decisions taken by two a
At this point, the last combinat

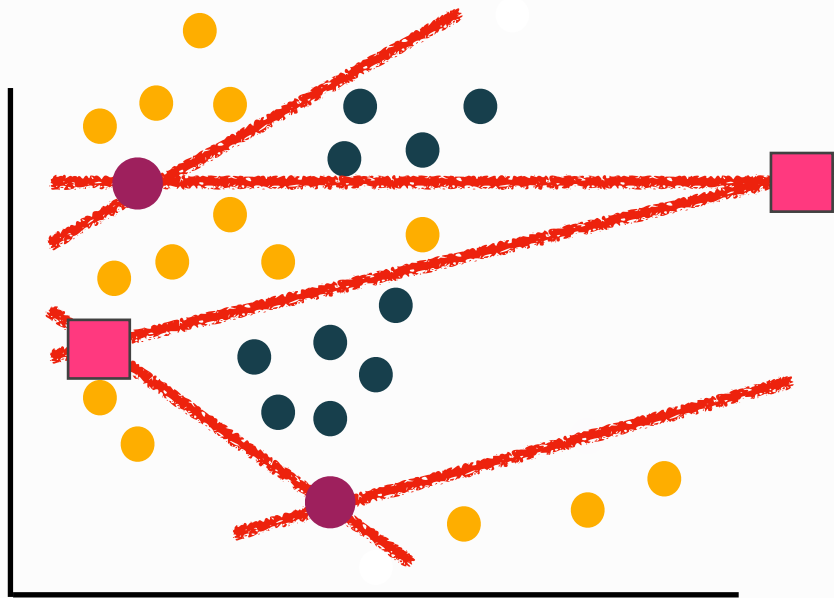
As we learnt, the first hidden layer is created by counting the number of lines: hence, for this example



Software Engineering for Artificial Intelligence: Configuring AI Solutions I

Configuring Artificial Intelligence Solutions: Advanced

Q8. How would you configure a deep learner given this distribution of data?



5 lines $\rightarrow$ 5 hidden neurons

Deeper is better. Many solutions here, but perhaps an additional layer may be enough.

Q9. How would you test your topology?

$$\text{Precision} = \frac{TP}{(TP + FP)}$$

$$\text{Recall} = \frac{TP}{(TP + FN)}$$

$$\text{Specificity} = \frac{TN}{(TN + FP)}$$

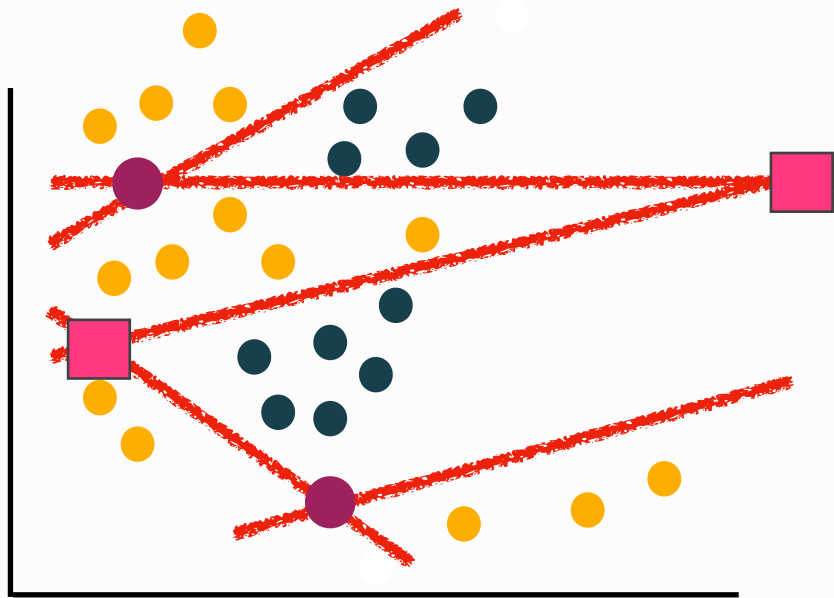
$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP) \cdot (TP + FN) \cdot (TN + FP) \cdot (TN + FN)}}$$

Software Engineering for Artificial Intelligence: Configuring AI Solutions I

Configuring Artificial Intelligence Solutions: Advanced

Q8. How would you configure a deep learner given this distribution of data?



5 lines $\rightarrow$ 5 hidden neurons

Deeper is better. Many solutions here, but perhaps an additional layer may be enough.

Q9. How would you test your topology?

Precision = $\frac{TP}{TP + FP}$

Be careful to accuracy! You may obtain a 99% accuracy without any true positive!

Specificity = $\frac{TP}{(TN + FP)}$

Accuracy = $\frac{(TP + TN)}{(TP + TN + FP + FN)}$

MCC = $\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP) \cdot (TP + FN) \cdot (TN + FP) \cdot (TN + FN)}}$

Software Engineering for Artificial Intelligence

Midterm In-Vivo Quiz

